AI Framework for Solar Power Generation Forecasting

Sai Neeraj
2420030197

Nalla Karthikeya
2420030161

C. Srikar
2420090025

M. Harshith Rao
2420090028

Course: **24ALT3101**

Team ID: **Team 17**

Institution: **KLH (Deemed to be University)**

Problem Statement

◆ Solar power output changes continuously because of sunlight, cloud cover, temperature, humidity and other weather conditions.

◆ Sudden changes in solar generation make grid balancing, battery scheduling and energy management more difficult.

◆ Traditional forecasting methods may struggle with nonlinear relationships between weather conditions and power output.

◆ The project proposes an AI-based framework that learns historical weather and generation patterns to forecast future solar power.

◆ Inaccurate forecasts can lead to poor energy planning, causing power wastage or shortages during changing weather conditions.

◆ A reliable forecasting system is needed to support better grid operation, battery storage planning, and efficient use of solar energy.

Objectives

- To develop an AI framework for short-term solar power generation forecasting.
- To develop a practical forecasting framework that can support real-time energy planning and efficient solar power utilization.
- To combine solar generation history with relevant weather and time-based features.
- To compare machine learning and deep learning models for forecasting performance.
- To reduce forecasting error and provide reliable predictions for energy management.
- To evaluate the framework using MAE, RMSE, MAPE and R^2.
- To identify the most important weather and environmental factors affecting solar power generation.

Literature

Research Paper	Techniques Used	Purpose	Limitation
Voyant et al. (2017) Machine learning methods for solar radiation forecasting	ANN, SVM, Random Forest, hybrid methods	Review and compare ML approaches for solar radiation forecasting.	Performance varies with location, inputs and data quality.
Bacher et al. (2009) Online short-term solar power forecasting	PV power history + weather information	Forecast near-term PV power using recent observations.	Rapid weather changes can reduce forecast accuracy.
Pedro & Coimbra (2012) Solar power production forecasting	Statistical and forecasting techniques	Compare techniques for short-term solar power prediction.	Results depend on forecast horizon and operating conditions.
Antonanzas et al. (2016) Review of photovoltaic power forecasting	Physical, statistical and ML approaches	Summarize major PV forecasting methods and applications.	No single method is best for every site and dataset.
Employee Attrition Prediction: An Explanatory and Statistically Robust Ensemble Learning Model — 2026	SHAP, LIME, Optuna, Random Oversampling, Ensemble Learning	To combine accurate prediction with global and local explanations and identify important retention factors.	Uses benchmark datasets, so real-world organizational validation remains important. (MDPI)

Research Gap Identification

- Site Dependence: A model trained for one solar plant may not perform equally well at another location.
- Weather Uncertainty: Cloud movement and sudden irradiance changes remain difficult to predict accurately.
- Limited Generalization: Many studies use one dataset or one forecasting horizon.
- Model Selection: ML and DL models behave differently depending on data size, features and forecast horizon.
- Operational Need: Forecasts should support grid scheduling, battery storage and energy planning—not only accuracy reporting.

Innovation

◆ **Hybrid AI Forecasting** – Combines machine learning/deep learning with weather and historical solar-generation data for improved prediction.
◆ **Real-Time Weather Integration** – Uses live irradiance, temperature, cloud cover, wind speed, and humidity data.
◆ **Adaptive Learning** – Continuously learns from new generation data and adapts to seasonal and changing weather patterns.
◆ **Multi-Time-Scale Prediction** – Provides **short-term, day-ahead, and long-term** solar power forecasts.
◆ **Uncertainty-Aware Forecasting** – Predicts not only generation but also a confidence range, helping grid operators manage uncertainty.

Creativity & Novelty

nformation in one forecasting pipeline.

- Novelty: Uses model comparison instead of depending on a single forecasting algorithm.
- Adaptive forecasting: The framework can be retrained when new plant data becomes available.
- Practical focus: Forecasts are designed for energy scheduling and storage decisions.
- Scalable design: The pipeline can be adapted to different solar plants and forecast horizons.
- Hybrid integration: Combines physical solar radiation dynamics with data-driven machine learning for higher accuracy.
- Uncertainty quantification: Provides confidence intervals alongside point forecasts to improve risk-managed grid operations.

Feasibility

support the workflow.

- Data feasibility: Public solar-generation and weather datasets can be used for academic experimentation.
- Implementation feasibility: Preprocessing, model training, evaluation and visualization can be developed in a notebook or dashboard.
- Evaluation feasibility: MAE, RMSE, MAPE, R^2 and forecast-error plots provide measurable performance indicators.
- Project scope: Suitable for an academic project and can later be extended to real-time forecasting.
- Cost feasibility: The project can be implemented using open-source AI/ML libraries and publicly available datasets, minimizing software and data costs.
- Time feasibility: The project can be completed within the academic timeline using a phased approach covering data preparation, model development, evaluation, and visualization.

Project Plan

- Phase 1 — Problem definition & requirement analysis
- Phase 2 — Dataset collection & preprocessing
- Phase 3 — Exploratory data analysis & feature engineering
- Phase 4 — ML/DL model development
- Phase 5 — Model comparison & hyperparameter tuning
- Phase 6 — Forecast evaluation & error analysis
- Phase 7 — Dashboard / visualization development
- Phase 8 — Testing, documentation & final presentation

Thankyou